Computational assessment of EBV-self peptide similarities across HLA class II contexts

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Abstract

Background Over the years, researchers have noticed that a prior Epstein-Barr virus (EBV) infection is present in almost all multiple sclerosis patients, with EBV seroconversion being followed by an increase in serological neurofilament light chain, a biomarker indicating neuroaxonal injury. Molecular mimicry between viral and self-antigens is one proposed mechanism for this association.

Results A multi-stage computational pipeline incorporating NetMHCIIpan, MixMHC2pred, nominally TCR-exposed BLOSUM62 similarity scoring, and structural modeling prioritized EBV-human peptide pairs for experimental peptide-HLA binding and register validation. Across 6,400 allele-specific comparisons in four HLA class II contexts, $\text{BALF5}_{627\text{-}641}$-$\text{TALDO1}_{108\text{-}122}$ (HLA-DRB1*13:03; rank 13/1,600) and $\text{BALF5}_{627\text{-}641}$-$\text{TALDO1}_{216\text{-}230}$ (HLA-DRB1*15:01; rank 14/1,600) were sequence-prioritized. In a separate enriched 49-pair evaluation panel, these were the only two pairs meeting the locked strict three-output, dual-arm binding/register gate at the 20% percentile cutoff; neither met that strict gate at 10% or below. The constrained sequence null did not yield a BH-adjusted $q < 0.05$ for any of the eight audited pairs, and both BALF5-TALDO1 comparisons had within-allele maxT $p = 1.0$. Structural models were sensitive to peptide input length and template choice, limiting register-specific interpretation. Exact-allele experimental binding evidence was identified only for the DRB1*15:01 viral arm.

Conclusions The study results provide a prioritized shortlist for experimental binding, register-mapping, and T-cell cross-reactivity testing.

1 Background

Multiple sclerosis (MS) is an autoimmune disease that can cause degeneration of motor, sensory, visual, and cognitive abilities through immune-mediated damage affecting myelin sheaths in the brain, spinal cord, and optic nerves [2]. In 2022, Bjornevik et al. studied a cohort of more than 10 million U.S. military personnel, 955 of whom developed MS. Their analysis reported an approximately 32-fold increase in MS risk following EBV infection; cytomegalovirus showed no comparable association. Serum neurofilament light chain increased only after EBV infection, indicating that EBV may precede detectable neurological injury [1]. Lanz et al. identified an antibody from an MS patient that recognized both EBV EBNA1 and the human CNS protein GlialCAM [3].

Previous studies such as Lanz et al. identified cross-reactive antibodies [3]; comprehensive mapping of T-cell cross-reactivity remains incompletely defined. T-cell receptors (TCRs) recognize a peptide presented in the binding groove of a human leukocyte antigen (HLA) molecule rather than the peptide alone. Sequence-only peptide comparisons therefore do not define the HLA-binding component of immune recognition. Different HLA alleles also show distinct structural preferences, affecting pMHC binding affinity and the binding register-the placement of the nine-amino-acid core within the HLA groove. The register determines which residues are anchors and which are potential TCR contacts [21, 22].

The computational pipeline developed here integrates peptide-HLA binding predictions, register-aware sequence alignments, and predicted peptide-HLA structures to prioritize a small subset of viral-self peptide pairs before laboratory testing. Predicted sequence or structural resemblance does not establish stable cell-surface presentation, functional T-cell activation in vivo, or experimentally confirmed registers.

2 Materials and Methods

2.1 Study design

This study describes a computational prioritization pipeline designed to identify high-interest EBV-human self-peptide candidate pairs for subsequent experimental peptide-HLA binding and binding-register validation. The workflow integrated predicted HLA-II presentation/binding support, register-aware sequence alignments, structural modeling, and downstream evidence review. Computational predictions alone do not establish natural processing and presentation in vivo, surface pMHC stability, or TCR recognition and functional cross-reactivity.

2.2 Peptide curation and allele-specific pairing

An EBV repertoire was curated from the Immune Epitope Database (IEDB) by filtering EBV source peptides with positive T-cell assay outcomes from MHC class II alleles in human hosts [4]. Human self-proteins included myelin-associated and CNS proteins from positive IEDB MHC-II T-cell records and additional 15-mer peptides derived from UniProt sequences [5]. Duplicate assay records sharing accession, coordinates, and sequence were merged. Imported IEDB records lacking an "Exact Epitope" classification or outside 11-30 amino acids were removed, as were sequences containing X, B, or Z. Forty EBV peptides and forty human peptides per allele, selected under evidence and protein-coverage rules, were paired within each allele to create 6,400 allele-specific comparisons.

The initial sequence screen was limited to HLA-DRB1*03:01, HLA-DRB1*08:01, HLA-DRB1*13:03, and HLA-DRB1*15:01. Each HLA allele was evaluated independently; pMHCs from different alleles were not compared. HLA-DRA*01:01 was the alpha chain in every pMHC model.

2.3 TCR-facing sequence-similarity scoring

The original 6,400-comparison V3 matrix contains one frozen, declared nine-residue core for each peptide arm. These declared cores were used as the input to the initial score; they were not changed after pairs were ranked. The local V2 register records identify the core caller as the IEDB recommended-binding service and retain the input peptide, allele, returned `core_peptide`, rank, IC50, and raw-response filename. The service's internal model version, invocation date, and any service-side tie behavior were not captured in the frozen response and cannot be reconstructed from the local files. The V3 protocol lock fixes the downstream ordering as TCR-facing BLOSUM62 score (descending), physicochemical mismatch (ascending), TCR-facing identity (descending), local-surface percentile (ascending), and lexical pair identifier (ascending). The initial ranking is therefore conditional on this declared-core input, while later predictor calls were used for audit and sensitivity analyses rather than to retrospectively replace a V3 core.

Sequence similarity was calculated with BLOSUM62 [8]. For cores a and b, at $I = \{\text{P2}, \text{P3}, \text{P5}, \text{P7}, \text{P8}\}$, the normalized score was $S(a,b) = \sum_{i \in I} B(a_i, b_i) \,/\, \sum_{i \in I} \max\{B(a_i, a_i), B(b_i, b_i)\}$. Thus, the denominator is the sum of the larger residue-specific self-substitution score at each compared position. Scores were calculated only for the nominally TCR-exposed/core non-anchor positions P2, P3, P5, P7, and P8 in aligned nine-residue binding cores. This custom normalization is bounded above by 1 for exact identity across the compared positions; lower

values reflect less favorable BLOSUM62 substitution support. Higher scores indicate greater substitution-matrix resemblance at this nominally exposed interface; they are not measures of T-cell recognition, cross-reactivity, presentation, or molecular mimicry. Percentile rankings were calculated independently within each allele's 1,600-pair cohort.

The two later lexicographic tie-break quantities were fixed, model-derived descriptors rather than additional biological evidence layers. Physicochemical mismatch summarizes scaled charge, hydropathy, hydrogen-bond donor/acceptor, and aromaticity differences at the nominally exposed residues. Local-surface percentile is the within-allele mean percentile of six peptide-surface features, each conservatively summarized by its 75th percentile across available model combinations: side-chain exposure mismatch, exposure-weighted centroid and orientation differences, exposed-residue distance-matrix difference, chemistry mismatch, and exposed-backbone RMSD. It is used only after the sequence keys and has no value when a structure is incomplete or fails the declared fixed-coordinate register-sensitivity rule (Supplementary Methods SM1).

2.4 Evidence audit and candidate prioritization

Eight candidate pairs selected from the 6,400-comparison matrix underwent a standardized manual evidence audit. Predicted binding support, predicted register agreement, IEDB sequence and provenance records, public immunopeptidome evidence, viral sequence conservation, and human self-protein tissue expression or rarity were evaluated. Pairs were classified as high priority, medium priority, or hold. No pair was high priority; two were medium priority; and six were held because one or both peptide arms did not meet binding criteria, provenance was unverified, immunopeptidome support was absent, or registers disagreed. The medium-priority pairs were BALF5$_{627\text{-}641}$-TALDO1$_{108\text{-}122}$ in HLA-DRB1*13:03 and BALF5$_{627\text{-}641}$-TALDO1$_{216\text{-}230}$ in HLA-DRB1*15:01. The eight-pair audit categories were separate from the later 49-pair provisional classifications.

2.5 Binding predictions and binding-register analysis

For the initial eight-pair review, peptide-HLA class II percentile ranks were predicted with NetMHCIIpan-4.3 EL and MixMHC2pred context predictions [6, 7]. These are presentation/eluted-ligand-oriented computational outputs, not direct equilibrium-affinity measurements. Full computational support required both predictors to return ranks of $\leq 5\%$ for both viral and self peptide arms. Lesser support required at least one predictor at $\leq 5\%$ or both predictors at $\leq 20\%$. Direct binding is described only when an exact peptide–allele assay record was identified.

Binding-register analysis defined the register as the predicted P1-P9 core within the open-ended HLA class II groove [21]. NetMHCIIpan-4.3 EL and MixMHC2pred context cores were compared. Any different core sequence, including shifted or overlapping nine-residue windows, was recorded as register disagreement. Major anchor positions (P1, P4, P6, P9) were separated from predicted TCR-facing positions (P2, P3, P5, P7, P8) for downstream analysis.

2.6 Expanded 49-pair cohort selection

The expanded 49-pair cohort was analyzed separately from the initial eight-pair evidence audit. All final pair identifiers matched entries in the original 6,400-comparison V3 matrix. The final cohort contained seven HLA-DRB1*03:01 pairs, 14 HLA-DRB1*08:01 pairs, 16 HLA-DRB1*13:03 pairs, and 12 HLA-DRB1*15:01 pairs, corresponding to 98 peptide arms. Thus, the expansion increased candidate coverage while retaining the same four alleles represented in the initial eight-pair audit (Supplementary Methods SM2).

2.7 Secondary provisional-tier rules

Secondary provisional tiers were assigned after the expanded 49-pair screen as routing labels for subsequent experimental binding and register-mapping work. No candidate received high-priority status because the automated public-evidence endpoint was unavailable under the frozen review protocol. These provisional labels do not establish peptide-HLA affinity, natural presentation, T-cell recognition, cross-reactivity, molecular mimicry, or disease relevance (Supplementary Methods SM2).

2.8 pMHC structural modeling

Heterotrimeric pMHC complexes (HLA-DRA, relevant HLA-DR beta chain, and target peptide) were modeled using the AlphaFold 3 web server, with five returned models per job [10]. Quality checks verified coordinate completion; peptide pLDDT and peptide-chain ipTM were reported separately because well-defined HLA chains can dominate global complex confidence. The first 85 Cα atoms of HLA chain A and chain B were aligned by Kabsch fitting. Peptide backbone RMSD was then measured over an 11-mer window nested around the proposed nine-residue core. AlphaFold outputs were treated as predicted models, not experimental coordinates, and analyzed independently of TCR-docking simulations.

2.9 PANDORA/MODELLER register stress test

PANDORA coupled with MODELLER was used as a cross-method, register-sensitivity stress test rather than as a binding-affinity or TCR-recognition model [17, 18]. Two preselected, forced pMHC-II templates were used: 1BX2 (primary) and 6CQQ (sensitivity template). For each template, only the first pMHC copy was retained and remapped to chains M/N/P; TCR chains and duplicate pMHC copies were excluded. Targets used HLA-DRA*01:01 and HLA-DRB1*15:01 sequences. Target termini were retained, rather than silently removed by template alignment. Each template–peptide–register condition requested 20 slow-loop-refinement models, with up to four local jobs. The local runtime was pinned to PANDORA commit `618bbb573645bd428697666475a2e47a2e418d01`; the locked environment record includes the MODELLER/PANDORA runtime probe. Reciprocal 1BX2/6CQQ calibration required complete 20-model runs, no target-template leakage, one expected-register top-five model below 2 Å, and a lower top-five median expected-register RMSD than both adjacent register decoys (Supplementary Methods SM8).

2.10 Calibration and exploratory modeling

The literature-derived Hy.2E11 cross-reactive benchmark, BALF5$_{627\text{-}641}$-MBP$_{85\text{-}99}$, served as a structural-comparison reference [11]. The benchmark contains HLA-DRA*01:01-HLA-DRB5*01:01 for BALF5 (PDB: 1H15) and HLA-DRA*01:01-HLA-DRB1*15:01 for MBP (PDB: 1BX2). It was therefore treated as a DR15 haplotype-level control, rather than a same-allele comparison; it does not validate either BALF5-TALDO1 pair or establish biological mimicry.

Poisson-Boltzmann/APBS pMHC surface-potential comparison was explored as an exploratory surface-resemblance descriptor [12]. Its score was excluded from primary ranking because a 30-result control gate yielded 15 failed and 12 missing results. Candidate outputs were excluded from ranking and cannot support shared-TCR or cross-reactivity claims; BALF5 replicate models remain exploratory internal predictions rather than docking validation (Supplementary Methods SM8).

2.11 Statistical calibration and sensitivity analyses

Statistical methods were implemented to test whether observed TCR-facing BLOSUM62 scores exceeded scores expected after disruption of residue-position alignments. Each HLA allele had its own separate simulation with a starting universe of 40 EBV cores and 40 self cores, making 1,600 pairwise comparisons. Ten thousand simulation replicates were completed independently for each allele at random seed `20260914` (Supplementary Methods SM3).

Next, threshold sensitivity of the frozen 49-pair panel was evaluated using percentile ranks from the IEDB-recommended binding method, NetMHCIIpan 4.3 EL, and MixMHC2pred 2.1 context. The specific percentile thresholds were 1%, 2%, 5%, 10%, and 20%, with exact-core and ± 1-residue register rules evaluated separately (Supplementary Methods SM3).

Four-metric rank sensitivity was also performed on the four allele-specific cohorts containing 1,600 pairs each. Pairs were evaluated using nominally TCR-exposed BLOSUM62, full-core BLOSUM62, nominally TCR-exposed sequence identity, and Grantham distance. A candidate was termed multi-metric rank-concordant when it was within the top 1% under at least three metrics; it was partially rank-concordant when it did not meet that criterion but was within the top 5% under at least three metrics (Supplementary Methods SM3).

2.12 Reproducibility

The local reproducibility package retains source files, software environments, random seeds, input logs, response-file references, and SHA-256 dataset checksums. Unevaluated or incomplete public-database evidence was labeled "Not Reported," rather than negative. No nominal p-values, FDR corrections, or probabilistic "mimicry probability" values were used in computational ranking. Public DOI archiving is intentionally deferred; a submission release would need to include the frozen source files, input manifests, commands, environment locks, and complete predictor outputs.

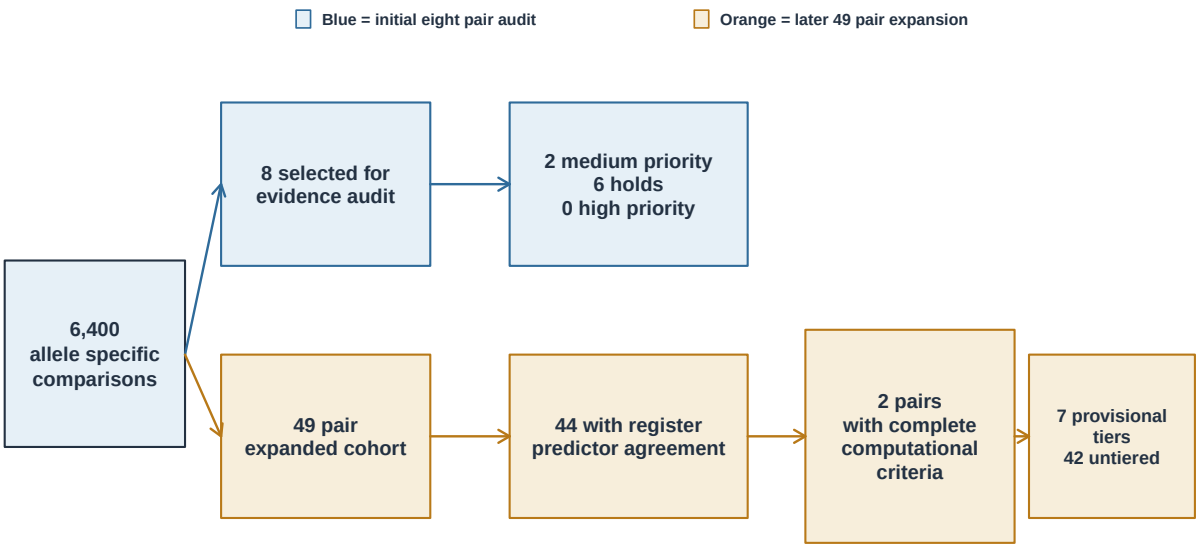


Figure 1: **Workflow for computational prioritization of EBV-human peptide pairs.** The 6,400 allele-specific comparisons informed an initial eight-pair evidence audit (blue) and a separate enriched 49-pair evaluation panel (orange). Boxes report the audit or provisional-classification outcomes for each track. The tracks used different review criteria and do not constitute a single sequential funnel or an independent validation sequence.

3 Results

3.1 Initial sequence comparisons across four HLA alleles

A total of 6,400 allele-specific peptide-pair comparisons were evaluated across four target alleles, with 1,600 pairs per allele. Sequence-similarity metrics were derived across nominally TCR-exposed/core non-anchor positions P2, P3, P5, P7, and P8. Actual TCR footprints can differ by receptor, allele, register, peptide conformation, and peptide-flanking-residue contacts. Score distributions were right-skewed across all alleles, with a small subset of pairs showing elevated nominally TCR-exposed sequence similarity (Supplementary Figure S1). These scores represent relative BLOSUM62 substitution-matrix resemblance, not percent identity or a probability of cross-reactivity. The top sequence-ranked pairs within each allele are reported in Supplementary Table S1.

BALF5$_{627\text{-}641}$-TALDO1$_{108\text{-}122}$ scored 0.28125 in HLA-DRB1*13:03 (rank 13/1,600), while BALF5$_{627\text{-}641}$-TALDO1$_{216\text{-}230}$ scored 0.31429 in HLA-DRB1*15:01 (rank 14/1,600). These are distinct TALDO1 peptide regions. Both had 40.0% identity across the five nominally TCR-exposed positions, with full-core identity of approximately 22.2% in *13:03 and 33.3% in *15:01.

The four-metric rank sensitivity was calculated for all 6,400 pairs (Figure 2). For the DR13 BALF5$_{627\text{-}641}$-TALDO1$_{108\text{-}122}$ pair, it met the multi-metric rank-concordance criterion with a nominally TCR-exposed BLOSUM62 rank of 13 (within the top 1%), nominally TCR-exposed identity rank of 2 (within the top 1%), a full-core BLOSUM62 rank of 173 (outside the top 5%), and a Grantham distance of 39.2, which is rank 15 (within the top 1%). For the DR15 BALF5$_{627\text{-}641}$-TALDO1$_{216\text{-}230}$ pair, it met the criterion with a nominally TCR-exposed BLOSUM62 rank of 14 (within the top 1%), a full-core BLOSUM62 rank of 3 (within the top 1%), a nominally TCR-exposed identity rank of 4 (within the top 1%), and a Grantham distance of 44.6, which is rank 33 (within

the top 5%). The pairwise within-allele Spearman correlation ρ-values ranged from 0.432 to 0.781, showing moderate-to-strong agreement among related sequence metrics. Competition ranks are used for this label, so ties can yield more than 16 pairs at the nominal top-1% boundary; the full rank and tie-size fields are retained in the V3 matrix. Although neither pair ranked uniformly highly across every metric, the DR13 pair was high in nominally TCR-exposed BLOSUM62, identity, and Grantham similarity, while the DR15 pair was high in both BLOSUM62 formulations and identity. These results remain prioritization results and are not cross-reactivity evidence.

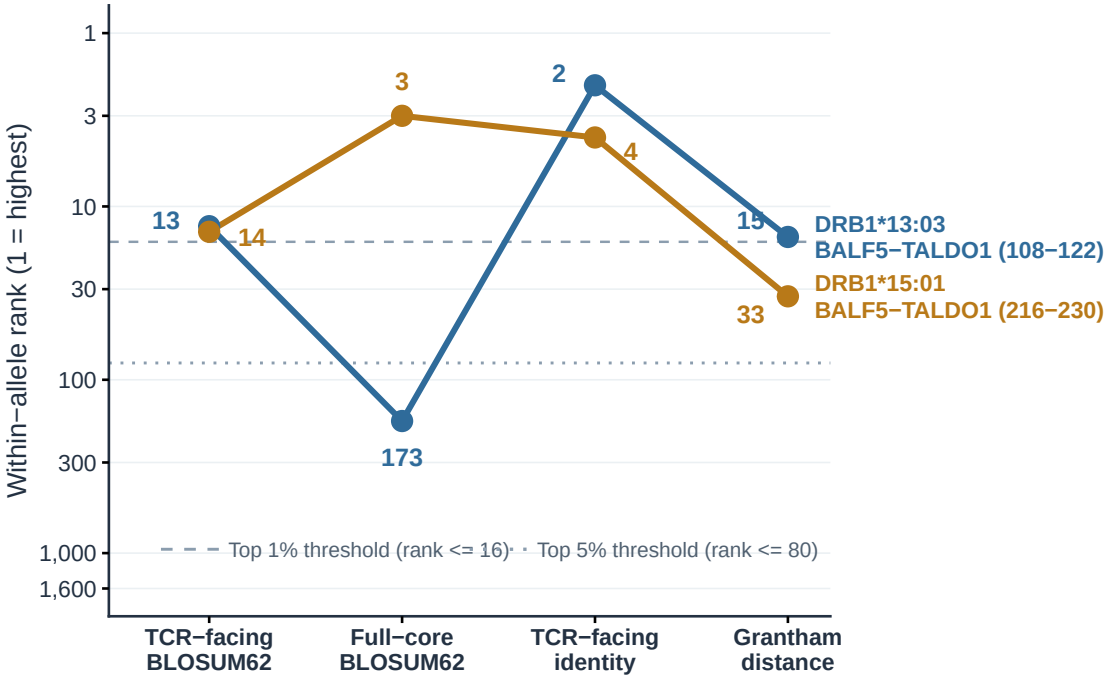


Figure 2: **Four-metric rank sensitivity of the BALF5-TALDO1 pairs.** Within-allele competition ranks are shown for TCR-facing BLOSUM62 similarity, full-core BLOSUM62 similarity, TCR-facing sequence identity, and Grantham distance. The vertical axis uses a reverse logarithmic scale, such that rank 1 indicates the greatest similarity. The HLA-DRB1*13:03 and HLA-DRB1*15:01 comparisons contain $\text{TALDO1}_{108\text{-}122}$ and $\text{TALDO1}_{216\text{-}230}$, respectively.

3.2 Constrained-null simulation results

Four of the eight audited pairs had nominal unadjusted p values below 0.05, but none met $q < 0.05$ after Benjamini-Hochberg adjustment (BH $q = 0.0632$-0.0819). Because the eight pairs were selected after inspection of the original screen, these BH values are descriptive post-selection calibration, not confirmatory discovery statistics. The within-allele maxT values are the more conservative family-wise summary; both BALF5-TALDO1 comparisons had maxT $p = 1.0$. Complete outputs, including null medians and 95% null intervals, are reported in Supplementary Table S4. This process tests only conditional sequence-score extremeness and does not provide information about presentation, TCR recognition, or cross-reactivity.

3.3 Initial Eight-Candidate Evidence Review

Eight candidates selected from the 6,400-pair matrix underwent deep evidence review, yielding two medium-priority candidates, six holds, and zero high-priority candidates (Supplementary Table S2). The two medium-

priority candidates were BALF5$_{627\text{-}641}$-TALDO1$_{108\text{-}122}$ in HLA-DRB1*13:03 and BALF5$_{627\text{-}641}$-TALDO1$_{216\text{-}230}$ in HLA-DRB1*15:01. Binding-prediction concordance is shown in Supplementary Figure S2.

Both BALF5-TALDO1 pairs showed predicted core-register agreement, although the registers remain unconfirmed experimentally. Neither pair satisfied full binding consensus. For the HLA-DRB1*13:03 candidate, the viral arm met the $\leq 5\%$ criterion while the self arm did not. In HLA-DRB1*15:01, neither arm met full-binding consensus. These thresholds describe the initial review; later predictor outputs differed. No corroborating experimental evidence under the review protocol, including IEDB eluted-ligand records or PRIDE-sourced immunopeptidome records, was identified for either self peptide [16]. The six held candidates had register discordance, insufficient binding support in one or both arms, or absent public experimental evidence. This review was an initial eight-pair evaluation and remained distinct from the later 49-pair provisional tiers.

3.4 Focused Structural Assessment of Predicted Binding Registers

The two medium-priority pairs were structurally assessed alongside a DRB5*01:01 restriction test. Thirteen AlphaFold 3 jobs, each returning five models, generated 65 structural models. All passed checks for coordinate completeness, correct chain identifiers, and peptide-sequence integrity. In HLA-DRB1*13:03, viral parent-nested median RMSD was 0.445 Å; BALF5 showed consistent coordinate recovery, whereas TALDO1$_{108\text{-}122}$ had lower confidence and positional uncertainty. In HLA-DRB1*15:01, viral and human RMSD values were 1.712 Å and 0.888 Å, respectively; viral BALF5 parent models varied, while the TALDO1$_{216\text{-}230}$ 11-mer was comparatively consistent. In HLA-DRB5*01:01, viral RMSD was 0.247 Å and longer parent-peptide modeling did not stabilize either competing TALDO1 register.

Models of shorter 11-mers frequently converged to consistent placements, but these placements were rarely retained when parent 15-mers were modeled, producing alternate poses and register shifts (Figure 3). Peptide-level pLDDT and peptide-chain ipTM were consequently examined separately from global pMHC metrics; HLA confidence can obscure peptide uncertainty. Sequence-predictor register agreement did not establish atomic-backbone structural confirmation.

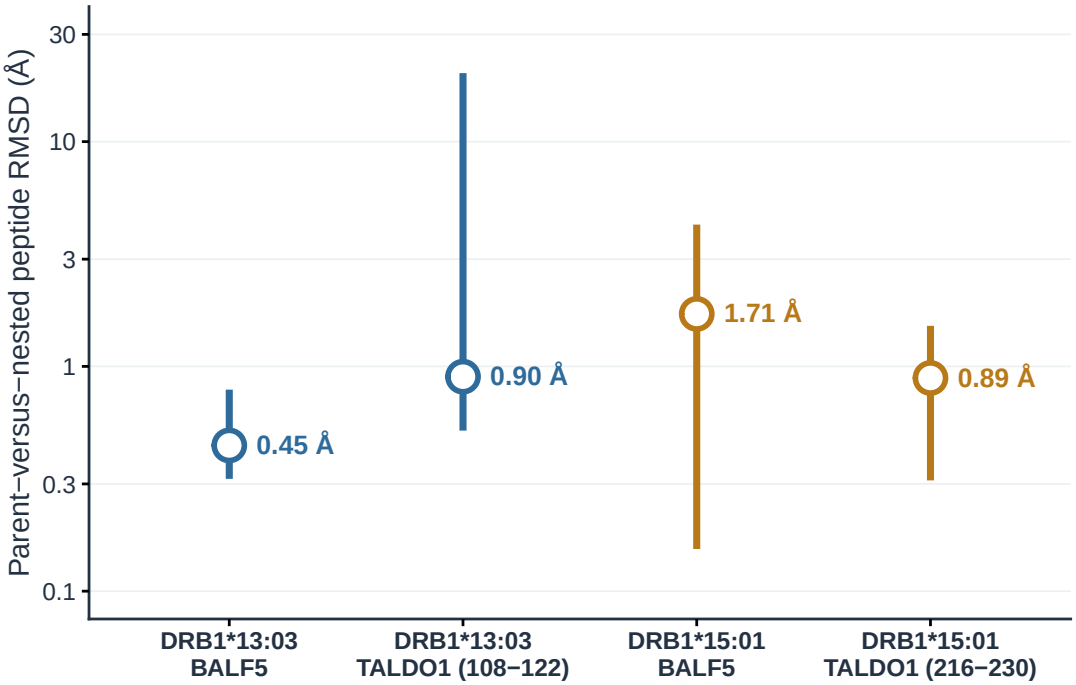


Figure 3: **Sensitivity of predicted peptide coordinates to input length.** Parent 15-mer and nested 11-mer models were compared for four peptide arms in HLA-DRB1*13:03 and HLA-DRB1*15:01. Vertical intervals show the minimum-maximum backbone RMSD across the 25 pairwise comparisons of five parent and five nested models; open circles show the median. The 25 comparisons are a descriptive 5×5 matrix, not 25 independent structural predictions. Values describe sensitivity of predicted coordinates to flanking sequence length and do not experimentally establish binding-register placement.

3.5 Cross-Method Structural Reproducibility

PANDORA coupled with MODELLER was used for cross-method structural reproducibility [17, 18]. The reciprocal PANDORA controls 1BX2 and 6CQQ were separate from the Hy.2E11 structural reference described above; both passed the pre-defined calibration gate: 120 models were assessed by expected-register recovery, adjacent-decoy comparison, sequence quality control, and template-leakage checks. Each control had at least one top-five model with expected-register core-backbone RMSD below 2 Å, lower top-five median RMSD than adjacent register decoys, and passed exact-sequence and template-leakage tests.

The subsequent ensemble contained 1,200 models spanning BALF5, TALDO1, and eight P6/P7 charge-substitution variants across two structural templates and three register placements (20 models per condition). None of the ten predefined peptides supported the expected NetMHCIIpan/MixMHC2pred register across both templates. A subsequent AlphaFold dataset comprised 150 models from 30 five-model jobs; seven charge-substitution comparisons had uncertain register placement and one was not evaluable because of seed mismatch. RMSD cannot on its own distinguish an integer register shift from a deformation within a register or from flanking-residue flexibility. Accordingly, these results identify structural uncertainty rather than a validated register state and do not support interpreting charge substitutions as evidence of disrupted binding.

3.6 Expanded Assessment of 49 Candidate Pairs

An enriched 49-pair evaluation panel was recovered from the 6,400-pair ranking; it is not an independent validation cohort. NetMHCIIpan, MixMHC2pred, and IEDB-recommended binding prediction profiles were completed for 98 peptide arms. Forty-four pairs had at least two outputs selecting the same declared nine-mer core for both arms; this is computational output concordance, not experimental register confirmation. Only $BALF5_{627\text{-}641}$-$TALDO1_{108\text{-}122}$ in HLA-DRB1*13:03 and $BALF5_{627\text{-}641}$-$TALDO1_{216\text{-}230}$ in HLA-DRB1*15:01 met the combined three-output binding/register criterion, with all three percentile ranks $\leq 20\%$ for both arms. This wording does not imply that the three outputs are independent biological measurements. Threshold sensitivity showed that this strict three-output result occurred only at the 20% cutoff; neither BALF5-TALDO1 pair passed it at 10% or below. Under a relaxed two-of-three-output rule, the DRB1*15:01 pair passed at 10%, whereas the DRB1*13:03 pair first passed at 20%. Allowing a ± 1-residue register tolerance added no pair under the strict rule. The two-pair result is therefore conditional on the permissive 20% gate rather than evidence of threshold-robust binding support.

For public-evidence assessment, 98 automated IEDB API queries returned zero retrievable records because the endpoint was unavailable. The resulting frozen status was not evaluable, so no candidate could be assigned high priority. Targeted manual IEDB verification later identified an experimentally confirmed binding record for the viral $BALF5_{627\text{-}641}$ peptide in HLA-DRB1*15:01 (IEDB Assay 1774674) [4, 20]. This evidence does not confirm $TALDO1_{216\text{-}230}$ binding, stable presentation, or dual-reactive T-cell recognition.

3.7 Secondary Provisional Classifications

A provisional shortlist contained seven candidates: one computational stratum A, two stratum B, and four stratum C. The remaining 42 candidates were untiered because evidence was insufficient for priority ranking. These labels are routing strata, not biological-confidence levels, and are distinct from the earlier eight-pair audit. The full provisional classification table is reported in Supplementary Table S3.

4 Discussion

4.1 Principal Findings

This multi-stage computational screening and evidence-review workflow prioritized EBV-human peptide pairs for later experimental validation. Its central findings were two BALF5-TALDO1 pairings, $BALF5_{627\text{-}641}$-$TALDO1_{216\text{-}230}$ and $BALF5_{627\text{-}641}$-$TALDO1_{108\text{-}122}$; despite sharing parent proteins, they are not the same epitope pair. The HLA-DRB1*15:01 pairing was assigned to computational stratum A, while the HLA-DRB1*13:03 pairing was assigned to stratum B. Secondary candidates included template-consistent $EBNA1_{87\text{-}101}$-$MBP_{217\text{-}233}$ in HLA-DRB1*15:01 and stratum-C candidates such as $EBNA2_{136\text{-}150}$-$MOG_{97\text{-}109}$ and $EBNA1_{87\text{-}101}$-$ANO2_{79\text{-}93}$. The outcome is a computational shortlist for in vitro peptide-HLA binding and register mapping, not a claim of functional molecular mimicry in vivo.

Computational stratum A does not mean "confirmed," validated, or "highest-confidence biological mimic."

4.2 Relevance of BALF5-TALDO1

BALF5-TALDO1 did not have the highest raw sequence-similarity score; EBNA1-ANO2 and EBNA3B-MOG scored higher. Raw sequence alignment was only the first evidence layer. The BALF5-TALDO1 protein pairing was represented by the only two allele-specific pairs in the 49-pair cohort that met combined computational binding and dual-arm register criteria. In HLA-DRB1*15:01, manual recovery of direct binding evidence for

viral BALF5$_{627\text{-}641}$ offered limited experimental grounding [4, 20]; the TALDO1 self arm remains unconfirmed by direct-binding or eluted-ligand assays. In HLA-DRB1*13:03, both arms met predictive thresholds, but no allele-specific public binding record was verified under the review protocol. The expanded provisional tiers do not replace the eight-pair audit categories; they derive from a later protocol with different criteria.

4.3 Contribution of Structural Analyses

Agreement among sequence-based register predictors did not guarantee stable structural coordinates. Shorter 11-mer inputs often converged in the models analyzed, whereas the same placements were often lost or shifted with the parent 15-mer input. Peptide-local pLDDT and peptide-chain ipTM were therefore more informative than global pMHC metrics, which can obscure peptide uncertainty through high HLA-chain confidence. AlphaFold 3 and PANDORA outputs showed register drift in the modeled BALF5/TALDO1 charge-substitution inputs. These analyses limit register-specific interpretation; they do not identify whether the observed uncertainty is a property of modeling conditions or of peptide biophysics. The retained EBNA1-MBP PANDORA register supports computational geometric plausibility, not native binding.

4.4 Linkage to Prior Research

Bjornevik et al. established the epidemiological link between EBV infection and MS, and Lanz et al. reported antibody cross-reactivity between EBNA1 and GlialCAM [1, 3]. These mechanisms differ from pMHC class II T-cell recognition; operationally, T-cell cross-reactivity requires one T cell to recognize more than one distinct peptide-MHC ligand [22]. Here, claims remain limited to computational candidate prioritization and predicted pMHC similarity. Literature-associated protein pairs, including EBNA1-ANO2 and EBNA1-MBP, emerged in the screen, but shared parent proteins must be distinguished from exact peptide epitopes [19]. Literature-informed pairings were not treated as independent computational validation because their disease association contributed to protein selection.

4.5 Additional Exploratory Methods

Electrostatic surface-potential comparison was excluded from primary ranking because 15 of 30 required control outputs failed and 12 were missing; among the three usable outputs, the calibration control did not discriminate from background decoys. TCRModel2 calibration produced acceptable models for 1ZGL but incorrect models for 1YMM and 2WBJ under DockQ and CAPRI criteria. These failures prevent candidate-specific TCR-recognition claims; they do not provide biological evidence that cross-reactivity is absent in vivo.

4.6 Limitations

The search space was limited to pre-curated viral epitopes, selected myelin/CNS proteins, and four HLA-DRB1 alleles. HLA-DQ and HLA-DP loci were omitted. Rankings shifted under TCR-facing BLOSUM62, full-core BLOSUM62, and Grantham-distance formulations. Predictors may share training data, potentially limiting the independence of their agreement. The unavailable public-evidence endpoint and absent self-peptide records did not establish biological absence of presentation. In silico modeling also cannot account for endogenous antigen processing, HLA-DM-mediated peptide editing, pMHC stability, antigen-presenting-cell context, or downstream TCR-signaling thresholds [23].

4.7 Experimental Next Steps

Peptide-HLA binding can be quantified for viral and self peptides against recombinant HLA-DRB1 molecules by fluorescence polarization or surface plasmon resonance. Register mapping may use alanine-scanning mutagenesis and structural methods such as X-ray crystallography or cryo-EM. Matched T cells should be tested against both peptide-HLA complexes for dual functional activation. Endogenous processing can be assessed through peptide elution and mass spectrometry from infected or inflamed antigen-presenting cells.

5 Conclusions

The first experimental priority is HLA-DRB1*15:01 BALF5$_{627\text{-}641}$-TALDO1$_{216\text{-}230}$, because it is in computational stratum A, passed the expanded binding-and-register gate at 20%, and has exact allele-specific viral-arm IEDB binding evidence [4, 20]. Recombinant peptide-HLA binding should test viral and self arms separately, followed by register mapping because structural analyses did not identify a single stable backbone register. The next candidate is HLA-DRB1*13:03 BALF5$_{627\text{-}641}$-TALDO1$_{108\text{-}122}$; both arms passed the expanded computational gate only at 20%, while absent public evidence was the major factor limiting its stratum. The third priority is HLA-DRB1*15:01 EBNA1$_{87\text{-}101}$-MBP$_{217\text{-}233}$, whose predicted register persisted across calibrated PANDORA template analysis despite failure of full binding consensus. The stratum-C candidates should follow once binding and register outcomes for these top three are available. Natural presentation should then be tested by immunopeptidomics, followed by matched-TCR assays against both pMHC complexes and functional-activation measurements; docking similarity alone is insufficient.

6 Abbreviations

AF3: AlphaFold 3; APBS: Adaptive Poisson-Boltzmann Solver; EBV: Epstein-Barr virus; HLA: human leukocyte antigen; IEDB: Immune Epitope Database; MS: multiple sclerosis; pMHC: peptide-major histocompatibility complex; RMSD: root-mean-square deviation; TCR: T-cell receptor.

7 Declarations

7.1 Ethics approval and consent to participate

Not applicable. This study used public database records and computational analyses and did not involve new research with human participants or animals.

7.2 Consent for publication

Not applicable.

7.3 Availability of data and materials

The analysis code, input manifests, derived data, and supplementary materials are available in the EBV-MS Project repository, `https://github.com/AnishSharma1/EBV-MS_Project`.

7.4 Competing interests

The author declares no competing interests.

7.5 Funding

No external funding was received for this work.

7.6 Authors' contributions

Anish Sharma developed the study, performed all analyses, and wrote the manuscript.

7.7 Acknowledgements

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